



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. Mohammed Razia Alangir Banu
Student Name	Thota Jhanasri
Roll Number	2520090121
Branch	CS-IT
Course Outcome	CO-1

Case Study Topic: Tourist Destination Indexing using AVL Tree

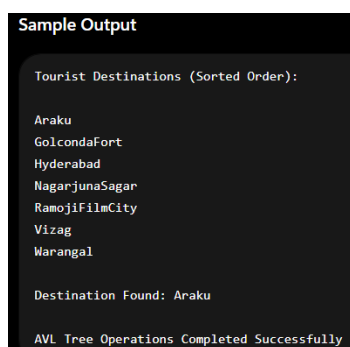
Work Done Summary:

In this case study, an AVL Tree was implemented for managing tourist destination records in the TourPlan – Smart Tourism & Travel Itinerary Optimization System. The AVL Tree stores destination names and maintains balance automatically after every insertion and deletion operation. Various balancing operations such as LL, RR, LR, and RL rotations were performed whenever the balance factor exceeded the allowed limit. The implementation demonstrates efficient searching, insertion, and deletion with $O(\log n)$ time complexity.

Implementation Details:

1. Created an AVL Tree node structure to store destination information.
2. Inserted tourist destinations into the AVL Tree.
3. Calculated height and balance factor for every node.
4. Applied LL, RR, LR, and RL rotations when required.
5. Implemented search operation for destination lookup.
6. Implemented deletion operation while maintaining tree balance.
7. Compared AVL Tree performance with a normal Binary Search Tree.
8. Verified that AVL Tree maintains $O(\log n)$ search performance.

RESULT : SCREENSHOTS/OUTPUT



This case study implements an AVL Tree for storing and managing tourist destinations in the TourPlan system. The AVL Tree maintains balance through rotations, ensuring efficient insertion, deletion, and search operations with $O(\log n)$ time complexity. The results demonstrate that AVL Trees provide faster and more reliable performance than normal BSTs for large tourism datasets.

GitHub link: https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

Student Signature	Faculty Signature